

MahalanobisDistance

Create the **MahalanobisDistance** function that we created in the lecture. It takes as input a single class from the pca model **mdl** created [1xN] feature vector **v**, containing N feature measurements for a sample. It outputs the Mahalanobis distance from the feature vector to the coordinates within the pca model space for the feature vector, and **std_per_mode**, which contains the standard deviation distances from the pca model.

The steps to perform in the function are as follows:

- compute **b**, our PCA coordinates for **v**, by subtracting **pcamd1.mu** from **v** and then projecting the result on multiplying by the eigenvectors matrix
- compute **std_per_mode**, the standard deviation distance of **v** from the mean for each mode of variation. This is normalized by the square root of the eigenvalues
- compute **md**, the Mahalanobis distance, as the square root of the sum of the squared entries in **std_per_mode**

Function ?



Save



Reset

MATLAB Documentation (<https://www.mathworks.com/help/>)

```

1 function [md,b,std_per_mode] = MahalanobisDistance(pcamdl,v)
2 % compute b, our PCA coordinates for v
3 b = pcamd1.eigvects*(v - pcamd1.mu)';
4 % compute D, the standard deviation distance from the mean for v for each mode of variation
5 std_per_mode = abs(b) ./ sqrt(pcamdl.eigvals);
6 % compute md, the Mahalanobis distance
7 md = sqrt(sum(std_per_mode.^2));
8 end

```

Code to call your function ?



Reset

```

1 mdl.class(1).eigvects = [0.7071 0.7071];
2 mdl.class(1).eigvals = 4;
3 mdl.class(1).mu = [2 2];
4 v = [2,2];
5 md = MahalanobisDistance(mdl.class(1),v) % md = 0 because v = mu
6 v = [2,2] + 2*[0.7071 0.7071];
7 md = MahalanobisDistance(mdl.class(1),v) % md = 1 because v is 2/sqrt(eigval)=2/2=1 standard deviations from mu
8 v = [2,2] - 6*[0.7071 0.7071];
9 md = MahalanobisDistance(mdl.class(1),v) % md = 3 because v is 6/sqrt(eigval)=6/2=3 standard deviations from mu
10 v = [2,2] + [-0.7071 0.7071];
11 md = MahalanobisDistance(mdl.class(1),v) % md = 0 because v lies in a direction from mu that is orthogonal

```

Run Function



Assessment: All Tests Passed

Submit



Is b correct?

Is std_per_mode correct?

Is md correct?

